

Initiating Search

November 5, 2025, 8:53 PM

• Search:

CAS Lexicon Search:

Weinreb amides - Preferred Concept

Search Tasks

Task	Result Type	View
Returned Reference Results from CAS Lexicon (143)	 References	View Results
Exported: Retrieved Related Reaction Results (579)	 Reactions	View Results

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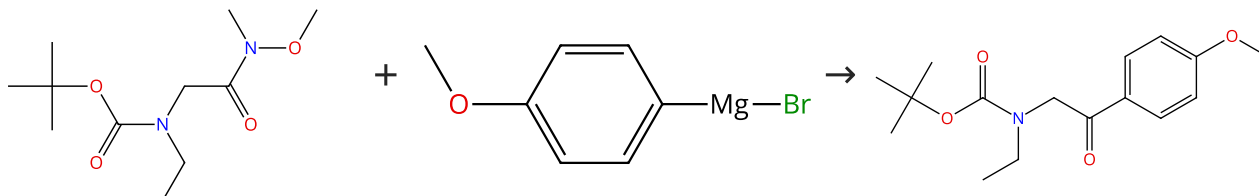
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Reactions (50)

[View in CAS SciFinder](#)

Scheme 1 (1 Reaction)

Steps: 1 Yield: 100%


 Suppliers (4)

 Suppliers (31)

 Supplier (1)

31-373-CAS-19473510

Steps: 1 Yield: 100%

Preparation of substituted 2-aminoimidazoles for the treatment of cancer

1.1 Solvents: Tetrahydrofuran; 1 h, -10 °C

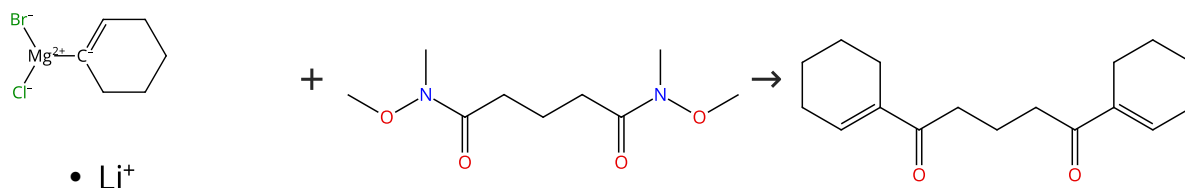
1.2 Reagents: Ammonium chloride
Solvents: Water

Assignees: Curza Global, LLC; The University of Utah

World Intellectual Property Organization, WO2018184019 A1
2018-10-04[PatentPak](#) available

Scheme 2 (1 Reaction)

Steps: 1 Yield: 100%

• Li⁺
 Suppliers (2)

 Supplier (1)

31-373-CAS-20702258

Steps: 1 Yield: 100%

Synthesis of α,ω -Bis-Enones by the Double Addition of Alkenyl Grignard Reagents to Diacid Weinreb Amides

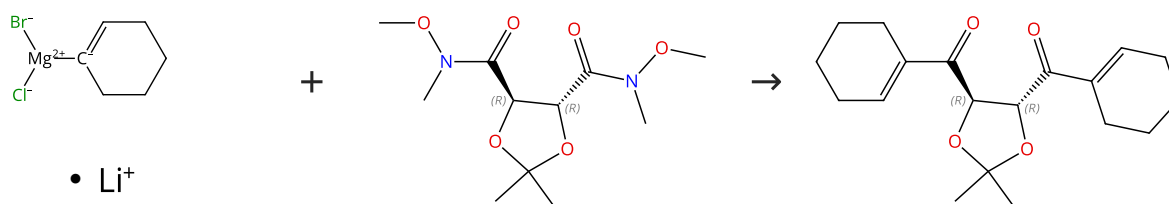
1.1 Solvents: Diethyl ether, Tetrahydrofuran; 10 min, 0 °C; 0 °C → rt; 14 - 20 h, rt

By: Wiesler, Stefan; et al

European Journal of Organic Chemistry (2019), 2019(36),
6246-6260.

Scheme 3 (1 Reaction)

Steps: 1 Yield: 100%

• Li⁺Absolute stereochemistry shown,
Rotation (-)

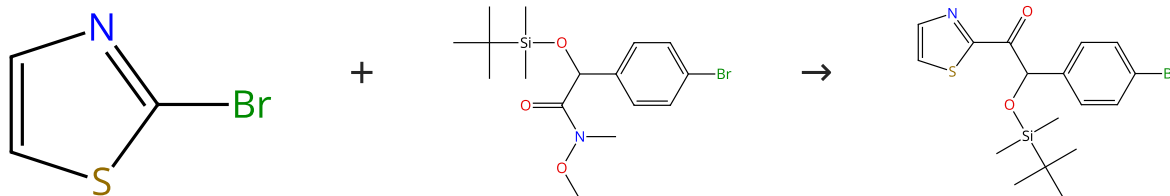
Absolute stereochemistry shown

 Suppliers (3)

31-373-CAS-20702264 Steps: 1 Yield: 100% 1.1 Solvents: Diethyl ether, Tetrahydrofuran; 10 min, 0 °C; 0 °C → rt; 14 - 20 h, rt	Synthesis of α,ω-Bis-Enones by the Double Addition of Alkenyl Grignard Reagents to Diacid Weinreb Amides By: Wiesler, Stefan; et al European Journal of Organic Chemistry (2019), 2019(36), 6246-6260.
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Scheme 4 (1 Reaction)

Steps: 1 Yield: 99%



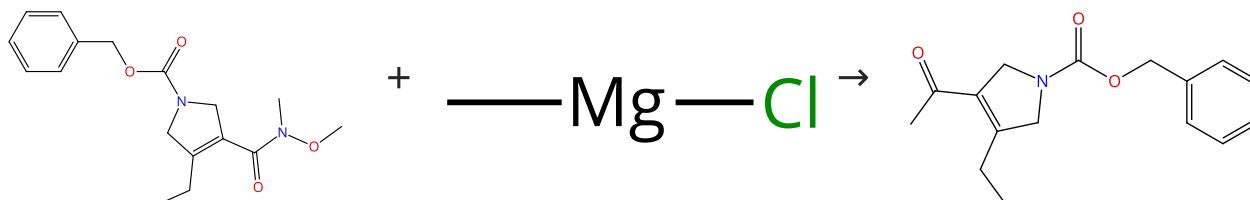
Suppliers (88)

Supplier (1)

31-614-CAS-32738510 Steps: 1 Yield: 99% 1.1 Reagents: Butyllithium Solvents: Tetrahydrofuran, Hexane; -78 °C; 1 h, -78 °C 1.2 Solvents: Tetrahydrofuran; -78 °C; 2.5 - 3 h, -78 °C 1.3 Reagents: Ammonium chloride Solvents: Water Experimental Protocols	Accessing Diverse Cross-Benzoin and α-Siloxy Ketone Products via Acyl Substitution Chemistry By: Larcombe, Chloe N.; et al Journal of Organic Chemistry (2022), 87(14), 9408-9413.
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Scheme 5 (1 Reaction)

Steps: 1 Yield: 98%

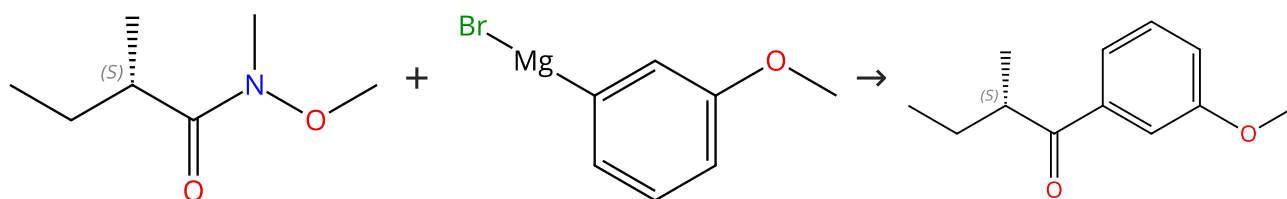


Suppliers (33)

31-373-CAS-23801167 Steps: 1 Yield: 98% 1.1 Solvents: Tetrahydrofuran; 0 °C; 0 °C → rt; 12 h, rt 1.2 Reagents: Ammonium chloride Solvents: Water	Process preparation of Upadacitinib and useful intermediates via Weinreb amides Assignee: Crystal Pharma, S.A.U. World Intellectual Property Organization, WO2021123288 A1 2021-06-24 PatentPak available
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Scheme 6 (1 Reaction)

Steps: 1 Yield: 97%

Absolute stereochemistry shown,
Rotation (+)

Absolute stereochemistry shown

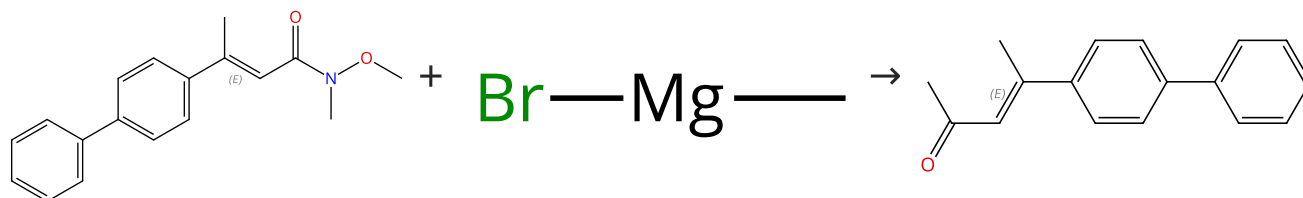
Suppliers (26)

Supplier (1)

31-614-CAS-24226806	Steps: 1 Yield: 97%	Stereodivergent Construction of Vicinal Acyclic Quaternary-Tertiary Carbon Stereocenters by Michael-Type Alkylation of α,α-Disubstituted N-tert-Butanesulfinyl Ketimines
1.1 Solvents: Tetrahydrofuran; 0 °C; 20 min, 0 °C; 0 °C → rt		By: Yisimayili, Nuermaimaiti; et al
1.2 Reagents: Ammonium chloride Solvents: Water		Organic Letters (2021), 23(19), 7450-7455.
Experimental Protocols		

Scheme 7 (1 Reaction)

Steps: 1 Yield: 97%



Double bond geometry shown

Suppliers (74)

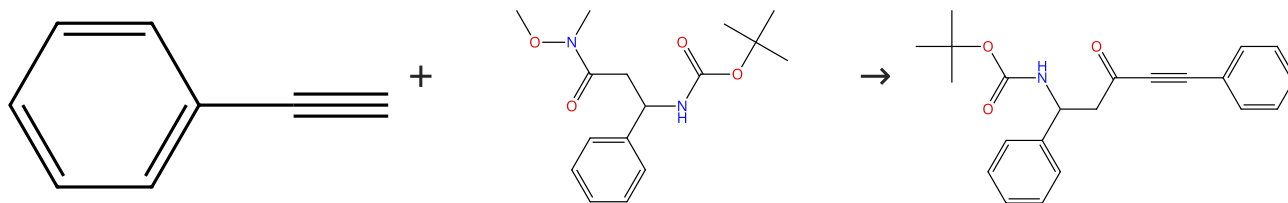
Double bond geometry shown

Suppliers (18)

31-373-CAS-18800530	Steps: 1 Yield: 97%	The synthesis of non-racemic β-alkyl-β-aryl-disubstituted allyl alcohols and their transformation into bearing a quaternary stereocenter allylamines and amino acids
1.1 Solvents: Tetrahydrofuran; -30 °C; -30 °C → -5 °C; 0.5 h, -5 °C		By: Narczyk, Aleksandra; et al
1.2 Reagents: Ammonium chloride Solvents: Water		Organic & Biomolecular Chemistry (2018), 16(21), 3921-3946.
Experimental Protocols		

Scheme 8 (1 Reaction)

Steps: 1 Yield: 97%



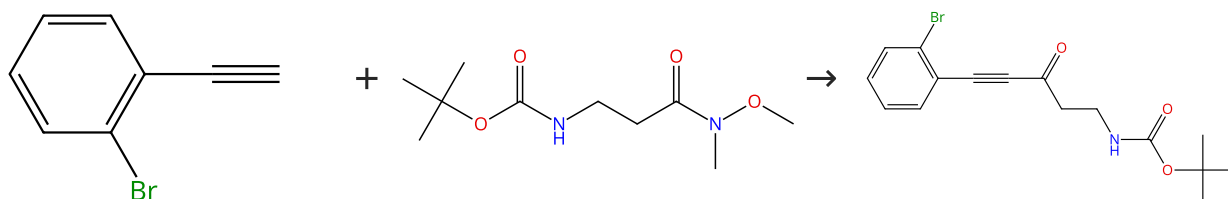
Suppliers (69)

Suppliers (2)

31-614-CAS-36832582	Steps: 1 Yield: 97%	Accessing spiropiperidines from dihydropyridones through tandem triflation-allylation and ring-closing metathesis (RCM)
1.1 Reagents: Butyllithium Solvents: Tetrahydrofuran, Hexane; -78 °C; 1 h, -78 °C		By: Gantasala, Naresh; et al
1.2 Solvents: Tetrahydrofuran; -78 °C; 15 min, -78 °C; 1 h, -50 °C; -50 °C → -5 °C; 1.5 h, -5 °C		Organic & Biomolecular Chemistry (2023), 21(25), 5245-5253.
1.3 Reagents: Monosodium phosphate Solvents: Water		
Experimental Protocols		

Scheme 9 (1 Reaction)

Steps: 1 Yield: 97%



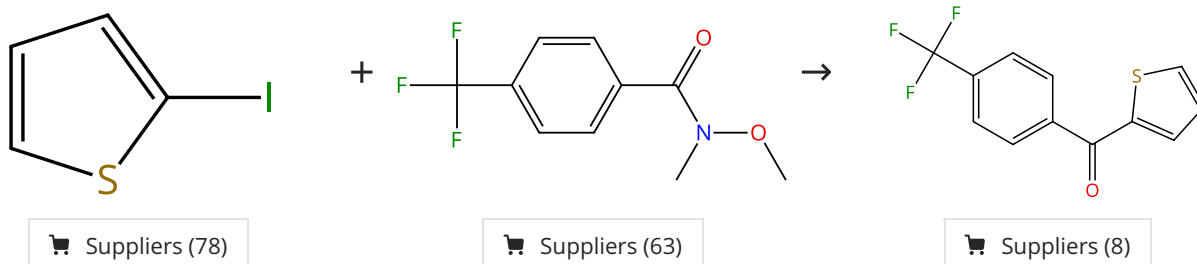
Suppliers (74)

Suppliers (52)

31-614-CAS-36832567	Steps: 1 Yield: 97%	Accessing spiropiperidines from dihydropyridones through tandem triflation-allylation and ring-closing metathesis (RCM)
1.1 Reagents: Butyllithium Solvents: Tetrahydrofuran, Hexane; -78 °C; 1 h, -78 °C		By: Gantasala, Naresh; et al
1.2 Solvents: Tetrahydrofuran; -78 °C; 15 min, -78 °C; 1 h, -50 °C; -50 °C → -5 °C; 1.5 h, -5 °C		Organic & Biomolecular Chemistry (2023), 21(25), 5245-5253.
1.3 Reagents: Monosodium phosphate Solvents: Water		
Experimental Protocols		

Scheme 10 (1 Reaction)

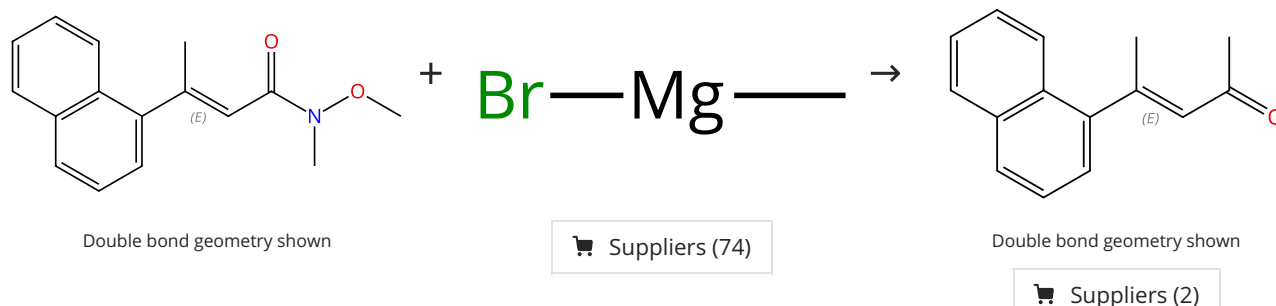
Steps: 1 Yield: 97%



31-373-CAS-22198696	Steps: 1 Yield: 97%	Synthesis of biaryl ketones by arylation of Weinreb amides with functionalized Grignard reagents under thermodynamic control vs. kinetic control of N,N-Boc ₂ -amides
1.1 Reagents: Magnesate(1-), dichloro(1-methylethyl)-, lithium (1:1) Solvents: Tetrahydrofuran; 30 min, 23 °C		By: Li, Guangchen; et al
1.2 Solvents: Toluene; 3 h, 23 °C		Organic & Biomolecular Chemistry (2020), 18(20), 3827-3831.
1.3 Reagents: Ammonium chloride Solvents: Water		
Experimental Protocols		

Scheme 11 (1 Reaction)

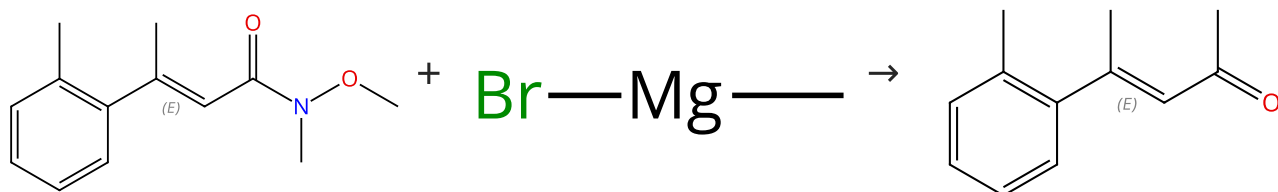
Steps: 1 Yield: 96%



31-373-CAS-18800531	Steps: 1 Yield: 96%	The synthesis of non-racemic β-alkyl-β-aryl-disubstituted allyl alcohols and their transformation into bearing a quaternary stereocenter allylamines and amino acids
1.1 Solvents: Tetrahydrofuran; -30 °C; -30 °C → -5 °C; 0.5 h, -5 °C		By: Narczyk, Aleksandra; et al
1.2 Reagents: Ammonium chloride Solvents: Water		Organic & Biomolecular Chemistry (2018), 16(21), 3921-3946.
Experimental Protocols		

Scheme 12 (1 Reaction)

Steps: 1 Yield: 96%



Double bond geometry shown

Supplier (1)

Suppliers (74)

Double bond geometry shown

Suppliers (2)

31-373-CAS-18800522

Steps: 1 Yield: 96%

1.1 Solvents: Tetrahydrofuran; -30 °C; -30 °C → -5 °C; 0.5 h, -5 °C

1.2 Reagents: Ammonium chloride
Solvents: Water

Experimental Protocols

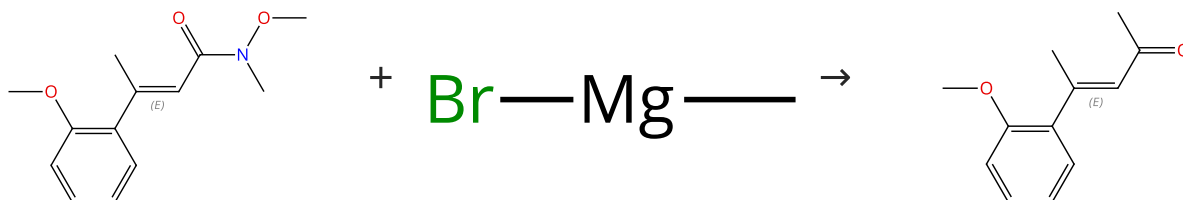
The synthesis of non-racemic β -alkyl- β -aryl-disubstituted allyl alcohols and their transformation into bearing a quaternary stereocenter allylamines and amino acids

By: Narczyk, Aleksandra; et al

Organic & Biomolecular Chemistry (2018), 16(21), 3921-3946.

Scheme 13 (1 Reaction)

Steps: 1 Yield: 96%



Double bond geometry shown

Suppliers (74)

Double bond geometry shown

Suppliers (2)

31-373-CAS-18800525

Steps: 1 Yield: 96%

1.1 Solvents: Tetrahydrofuran; -30 °C; -30 °C → -5 °C; 0.5 h, -5 °C

1.2 Reagents: Ammonium chloride
Solvents: Water

Experimental Protocols

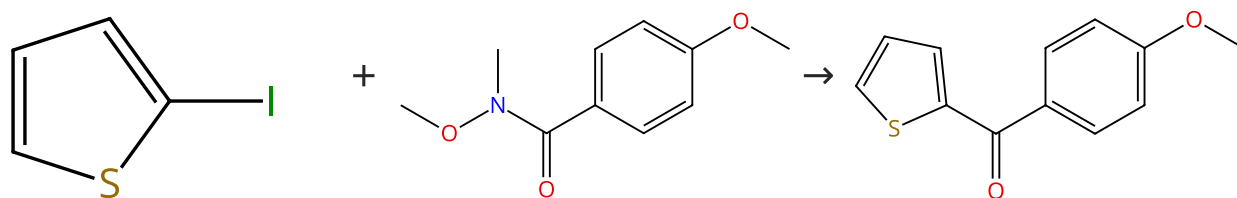
The synthesis of non-racemic β -alkyl- β -aryl-disubstituted allyl alcohols and their transformation into bearing a quaternary stereocenter allylamines and amino acids

By: Narczyk, Aleksandra; et al

Organic & Biomolecular Chemistry (2018), 16(21), 3921-3946.

Scheme 14 (1 Reaction)

Steps: 1 Yield: 96%



Suppliers (78)

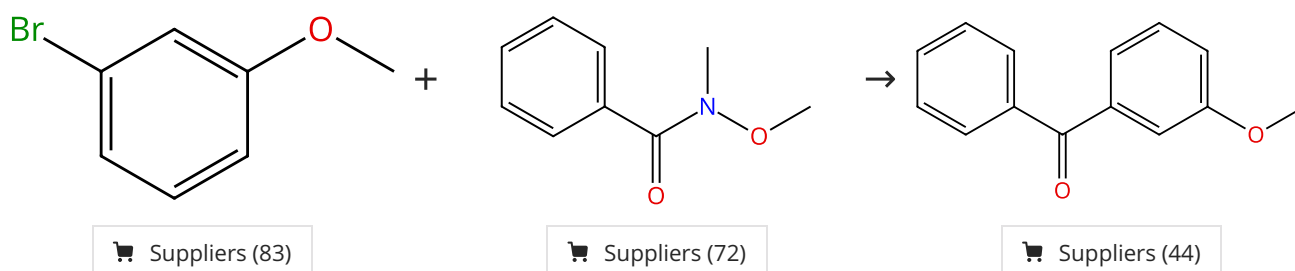
Suppliers (65)

Suppliers (77)

31-373-CAS-22198695	Steps: 1 Yield: 96%	Synthesis of biaryl ketones by arylation of Weinreb amides with functionalized Grignard reagents under thermodynamic control vs. kinetic control of N,N-Boc₂-amides
1.1 Reagents: Magnesate(1-), dichloro(1-methylethyl)-, lithium (1: 1) Solvents: Tetrahydrofuran; 30 min, 23 °C		By: Li, Guangchen; et al
1.2 Solvents: Toluene; 3 h, 23 °C		Organic & Biomolecular Chemistry (2020), 18(20), 3827-3831.
1.3 Reagents: Ammonium chloride Solvents: Water		
Experimental Protocols		

Scheme 15 (1 Reaction)

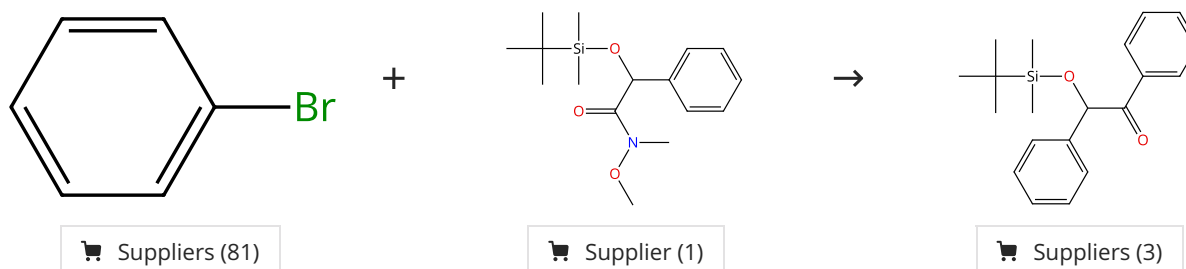
Steps: 1 Yield: 96%



31-373-CAS-22198674	Steps: 1 Yield: 96%	Synthesis of biaryl ketones by arylation of Weinreb amides with functionalized Grignard reagents under thermodynamic control vs. kinetic control of N,N-Boc₂-amides
1.1 Reagents: Magnesate(1-), dichloro(1-methylethyl)-, lithium (1: 1) Solvents: Tetrahydrofuran; 3 d, 23 °C		By: Li, Guangchen; et al
1.2 Solvents: Toluene; 3 h, 23 °C		Organic & Biomolecular Chemistry (2020), 18(20), 3827-3831.
1.3 Reagents: Ammonium chloride Solvents: Water		
Experimental Protocols		

Scheme 16 (1 Reaction)

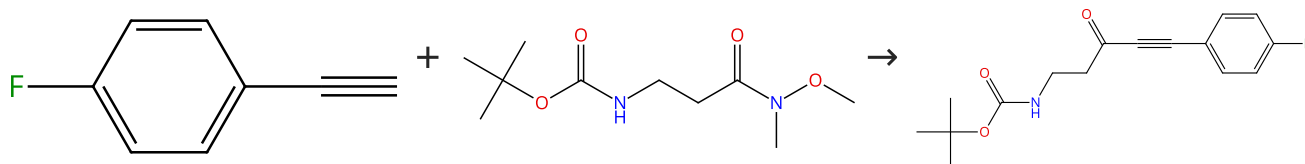
Steps: 1 Yield: 96%



31-614-CAS-32738492	Steps: 1 Yield: 96%	Accessing Diverse Cross-Benzoin and α-Siloxy Ketone Products via Acyl Substitution Chemistry
1.1 Reagents: Butyllithium Solvents: Tetrahydrofuran, Hexane; -78 °C; 1 h, -78 °C		By: Larcombe, Chloe N.; et al
1.2 Solvents: Tetrahydrofuran; -78 °C; 2.5 - 3 h, -78 °C		Journal of Organic Chemistry (2022), 87(14), 9408-9413.
1.3 Reagents: Ammonium chloride Solvents: Water		
Experimental Protocols		

Scheme 17 (1 Reaction)

Steps: 1 Yield: 96%



Suppliers (74)

Suppliers (52)

Supplier (1)

31-614-CAS-36832572

Steps: 1 Yield: 96%

Accessing spiropiperidines from dihydropyridones through tandem triflation-allylation and ring-closing metathesis (RCM)

By: Gantasala, Naresh; et al

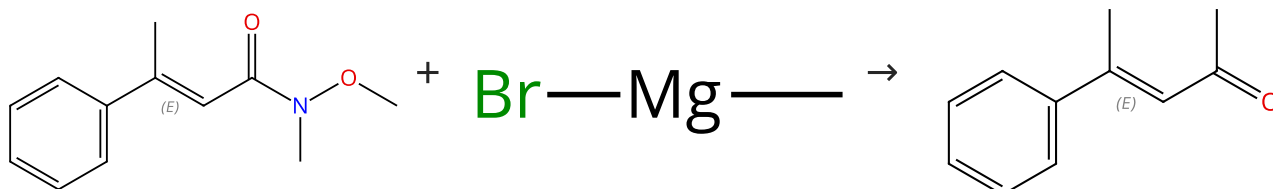
Organic & Biomolecular Chemistry (2023), 21(25), 5245-5253.

- 1.1 **Reagents:** Butyllithium
Solvents: Tetrahydrofuran, Hexane; -78 °C; 1 h, -78 °C
- 1.2 **Solvents:** Tetrahydrofuran; -78 °C; 15 min, -78 °C; 1 h, -50 °C; -50 °C → -5 °C; 1.5 h, -5 °C
- 1.3 **Reagents:** Monosodium phosphate
Solvents: Water

Experimental Protocols

Scheme 18 (1 Reaction)

Steps: 1 Yield: 96%



Double bond geometry shown

Suppliers (2)

Suppliers (74)

Double bond geometry shown

Suppliers (3)

31-373-CAS-18800521

Steps: 1 Yield: 96%

The synthesis of non-racemic β -alkyl- β -aryl-disubstituted allyl alcohols and their transformation into bearing a quaternary stereocenter allylamines and amino acids

By: Narczyk, Aleksandra; et al

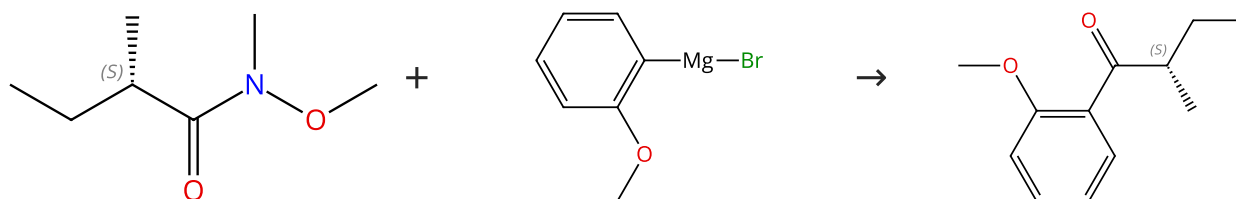
Organic & Biomolecular Chemistry (2018), 16(21), 3921-3946.

- 1.1 **Solvents:** Tetrahydrofuran; -30 °C; -30 °C → -5 °C; 0.5 h, -5 °C
- 1.2 **Reagents:** Ammonium chloride
Solvents: Water

Experimental Protocols

Scheme 19 (1 Reaction)

Steps: 1 Yield: 96%

Absolute stereochemistry shown,
Rotation (+)

Supplier (1)

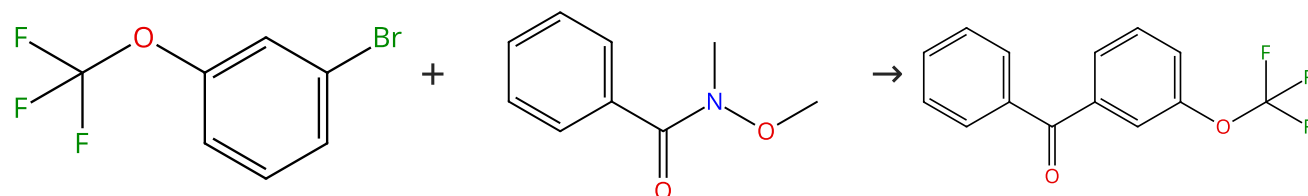
Suppliers (25)

Absolute stereochemistry shown

31-614-CAS-24226820	Steps: 1 Yield: 96%	Stereodivergent Construction of Vicinal Acyclic Quaternary-Tertiary Carbon Stereocenters by Michael-Type Alkylation of α,α-Disubstituted N-tert-Butanesulfinyl Ketimines
1.1 Solvents: Tetrahydrofuran; 0 °C; 20 min, 0 °C; 0 °C → rt		By: Yisimayili, Nuermaimaiti; et al
1.2 Reagents: Ammonium chloride Solvents: Water		Organic Letters (2021), 23(19), 7450-7455.
Experimental Protocols		

Scheme 20 (1 Reaction)

Steps: 1 Yield: 95%



Suppliers (78)

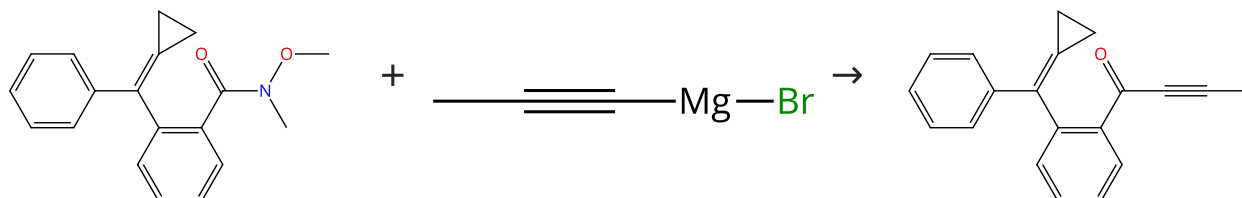
Suppliers (72)

Suppliers (10)

31-373-CAS-22198675	Steps: 1 Yield: 95%	Synthesis of biaryl ketones by arylation of Weinreb amides with functionalized Grignard reagents under thermodynamic control vs. kinetic control of N,N-Boc₂-amides
1.1 Reagents: Magnesate(1-), dichloro(1-methylethyl)-, lithium (1: 1) Solvents: Tetrahydrofuran; 2 d, 23 °C		By: Li, Guangchen; et al
1.2 Solvents: Toluene; 3 h, 23 °C		Organic & Biomolecular Chemistry (2020), 18(20), 3827-3831.
1.3 Reagents: Ammonium chloride Solvents: Water		
Experimental Protocols		

Scheme 21 (1 Reaction)

Steps: 1 Yield: 95%

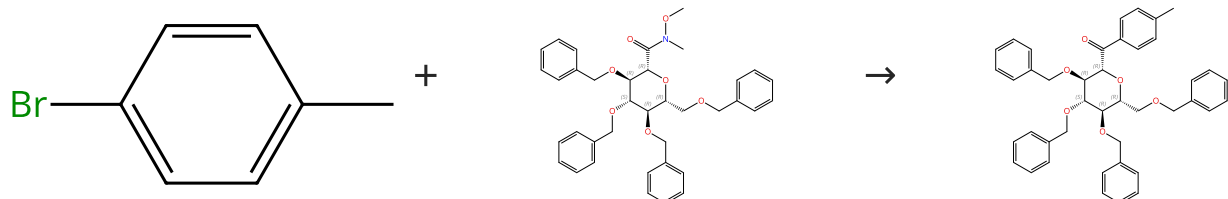


Suppliers (17)

31-614-CAS-34705314	Steps: 1 Yield: 95%	Construction of Cyclopenta[b]naphthalenol Frameworks by Weinreb Amide-Directed Umpolung Ring Opening of Methylenecyclopropanes
1.1 Solvents: Tetrahydrofuran; rt; 1.5 h, 70 °C; 70 °C → 0 °C		By: Wei, Hao-Zhao; et al
1.2 Reagents: Water; rt		Advanced Synthesis & Catalysis (2022), 364(22), 3948-3953.
Experimental Protocols		

Scheme 22 (1 Reaction)

Steps: 1 Yield: 95%



Suppliers (67)

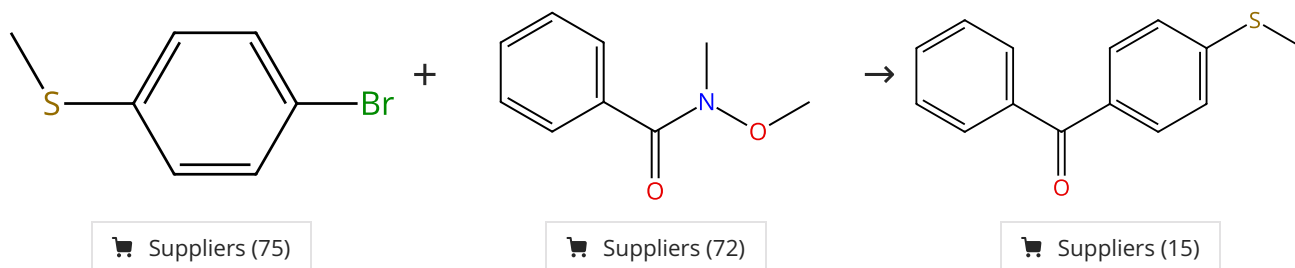
Absolute stereochemistry shown

Absolute stereochemistry shown

31-373-CAS-20640561 Steps: 1 Yield: 95%	Acyl and Benzyl-C-β-D-Glucosides: Synthesis and Biostudies for Glucose-Uptake-Promoting Activity in C2 C12 Mytotubes
1.1 Reagents: Magnesium Catalysts: Iodine Solvents: Tetrahydrofuran; rt; rt → 0 °C 1.2 Solvents: Tetrahydrofuran; 2 h, 0 °C	By: Reddy, Mannem Rajeswara; et al European Journal of Organic Chemistry (2019), 2019(35), 6053-6070.

Scheme 23 (1 Reaction)

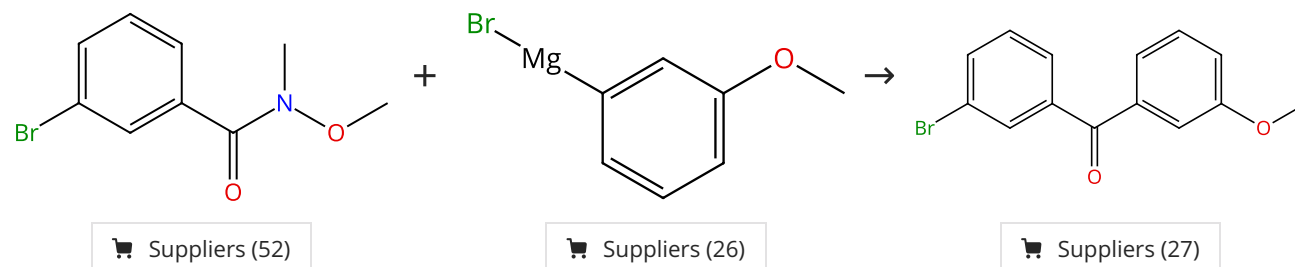
Steps: 1 Yield: 95%



31-373-CAS-22198663 Steps: 1 Yield: 95%	Synthesis of biaryl ketones by arylation of Weinreb amides with functionalized Grignard reagents under thermodynamic control vs. kinetic control of N,N-Boc₂-amides
1.1 Reagents: Magnesate(1-), dichloro(1-methylethyl)-, lithium (1: 1) Solvents: Tetrahydrofuran; 2 d, 23 °C 1.2 Solvents: Toluene; 3 h, 23 °C 1.3 Reagents: Ammonium chloride Solvents: Water Experimental Protocols	By: Li, Guangchen; et al Organic & Biomolecular Chemistry (2020), 18(20), 3827-3831.

Scheme 24 (1 Reaction)

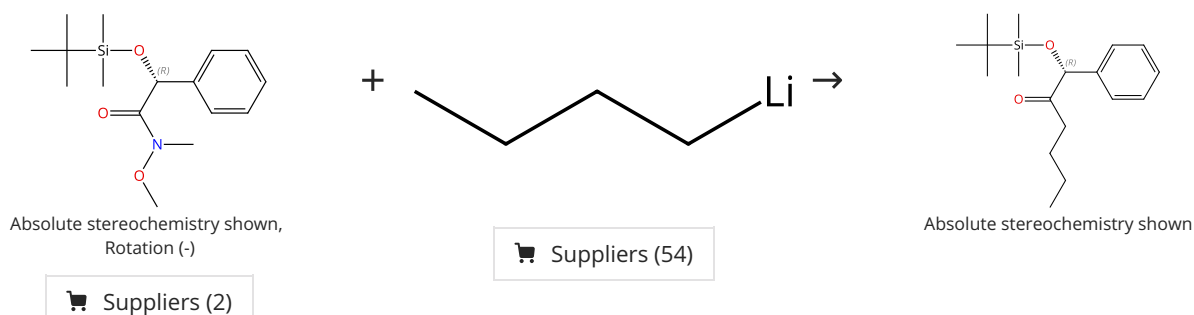
Steps: 1 Yield: 95%



31-614-CAS-25042618 Steps: 1 Yield: 95%	An efficient and concise synthesis of a selective small molecule non-peptide inhibitor of cathepsin L: KGP94
1.1 Solvents: Tetrahydrofuran; 30 min, 0 °C; 0 °C → rt; 4 h, rt 1.2 Reagents: Water; 0 °C Experimental Protocols	By: Munikishore, Rachakunta; et al Bioorganic Chemistry (2021), 116, 105317.

Scheme 25 (1 Reaction)

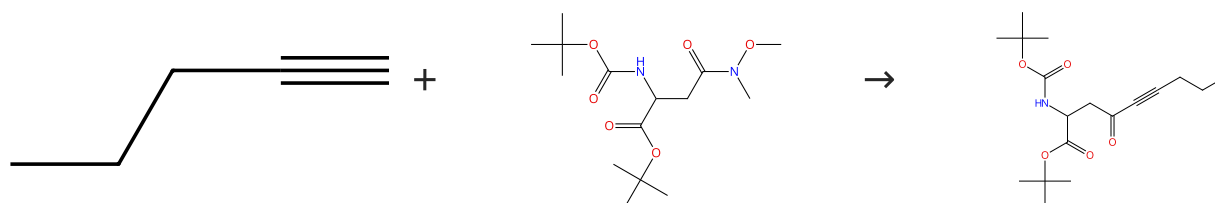
Steps: 1 Yield: 95%



31-614-CAS-32738514 Steps: 1 Yield: 95% 1.1 Solvents: Tetrahydrofuran, Hexane; -78 °C; 2.5 h, -78 °C 1.2 Reagents: Ammonium chloride Solvents: Water Experimental Protocols	Accessing Diverse Cross-Benzoin and α-Siloxy Ketone Products via Acyl Substitution Chemistry By: Larcombe, Chloe N.; et al Journal of Organic Chemistry (2022), 87(14), 9408-9413.
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Scheme 26 (1 Reaction)

Steps: 1 Yield: 95%

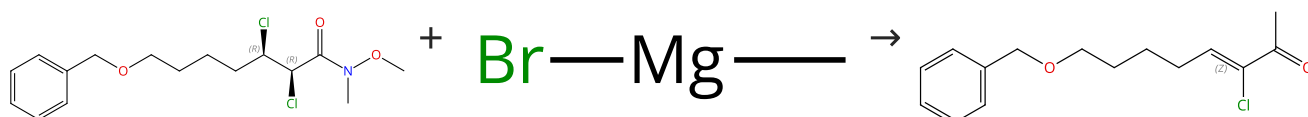


Suppliers (42)

31-614-CAS-36832579 Steps: 1 Yield: 95% 1.1 Reagents: Butyllithium Solvents: Tetrahydrofuran, Hexane; -78 °C; 1 h, -78 °C 1.2 Solvents: Tetrahydrofuran; -78 °C; 15 min, -78 °C; 1 h, -50 °C; -50 °C → -5 °C; 1.5 h, -5 °C 1.3 Reagents: Monosodium phosphate Solvents: Water Experimental Protocols	Accessing spiropiperidines from dihydropyridones through tandem triflation-allylation and ring-closing metathesis (RCM) By: Gantasala, Naresh; et al Organic & Biomolecular Chemistry (2023), 21(25), 5245-5253.
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Scheme 27 (1 Reaction)

Steps: 1 Yield: 94%



Relative stereochemistry shown

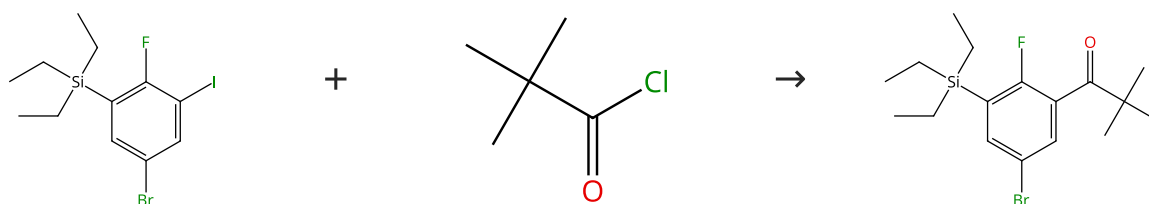
Suppliers (74)

Double bond geometry shown

31-614-CAS-24044182 Steps: 1 Yield: 94% 1.1 Solvents: Tetrahydrofuran; 120 min, 0 °C 1.2 Reagents: Hydrochloric acid Solvents: 1,4-Dioxane; 0 °C Experimental Protocols	Efficient synthesis of α,β-dichlorinated ketones from α,β-dichlorinated Weinreb amides through a simple work-up procedure By: Prakoso, Nurcahyo Iman; et al Organic & Biomolecular Chemistry (2021), 19(36), 7822-7826.
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Scheme 28 (1 Reaction)

Steps: 1 Yield: 94%

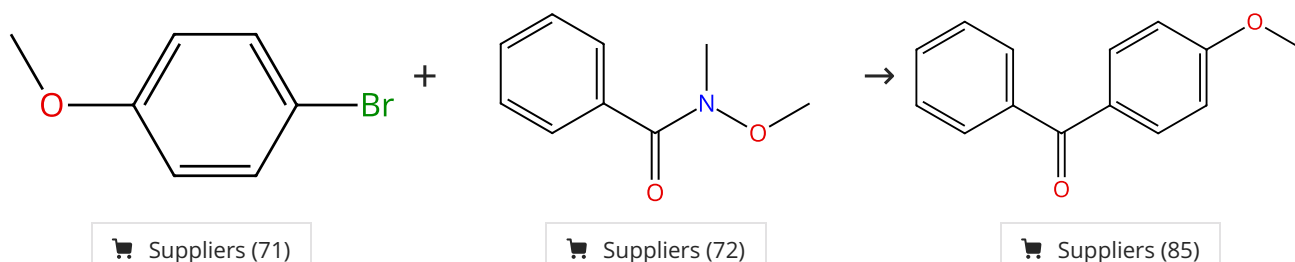


Suppliers (42)

31-373-CAS-19645928	Steps: 1 Yield: 94%	Highly Regioselective Remote Lithiation of Functionalized 1,3-bis-Silylated Arenes
1.1 Catalysts: Cuprous cyanide, Lithium chloride; 5 h, 150 °C; 150 °C → 0 °C		By: Bellan, Andreas B.; et al
1.2 Solvents: Tetrahydrofuran; 0 °C		Angewandte Chemie, International Edition (2019), 58(6), 1838-1841.
1.3 Reagents: Magnesate(1-), dichloro(1-methylethyl)-, lithium (1:1)		
Solvents: Tetrahydrofuran; -40 °C; 15 min, -40 °C		
1.4 Solvents: Tetrahydrofuran; -40 °C → 25 °C		
1.5 Reagents: Ammonium chloride		
Solvents: Water		
Experimental Protocols		

Scheme 29 (1 Reaction)

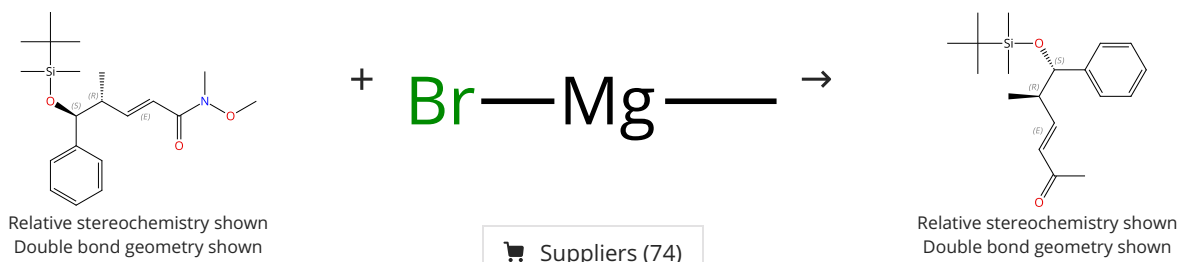
Steps: 1 Yield: 94%



31-373-CAS-22198662	Steps: 1 Yield: 94%	Synthesis of biaryl ketones by arylation of Weinreb amides with functionalized Grignard reagents under thermodynamic control vs. kinetic control of N,N-Boc₂-amides
1.1 Reagents: Magnesate(1-), dichloro(1-methylethyl)-, lithium (1:1)		By: Li, Guangchen; et al
Solvents: Tetrahydrofuran; 3 d, 23 °C		Organic & Biomolecular Chemistry (2020), 18(20), 3827-3831.
1.2 Solvents: Toluene; 3 h, 23 °C		
1.3 Reagents: Ammonium chloride		
Solvents: Water		
Experimental Protocols		

Scheme 30 (1 Reaction)

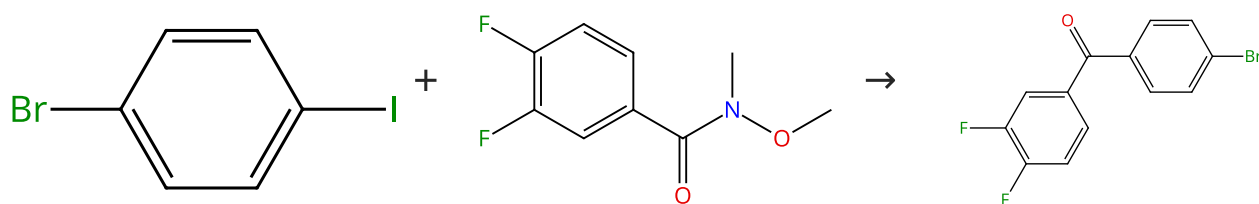
Steps: 1 Yield: 94%



31-614-CAS-43179293	Steps: 1 Yield: 94%	(E)-selective Weinreb amide-type Horner-Wadsworth-Emmons reaction: effect of reaction conditions, substrate scope, isolation of a reactive magnesium phosphonoenolate, and applications
1.1 Solvents: Tetrahydrofuran; 10 min, -78 °C; 30 min, 0 °C		By: Murata, Takatsugu; et al
1.2 Reagents: Ammonium chloride		ChemRxiv (2024), 1-26.
Solvents: Water; 0 °C		
Experimental Protocols		

Scheme 31 (1 Reaction)

Steps: 1 Yield: 94%



Suppliers (92)

Suppliers (38)

Suppliers (32)

31-373-CAS-22198699

Steps: 1 Yield: 94%

Synthesis of biaryl ketones by arylation of Weinreb amides with functionalized Grignard reagents under thermodynamic control vs. kinetic control of N,N-Boc₂-amides

By: Li, Guangchen; et al

Organic & Biomolecular Chemistry (2020), 18(20), 3827-3831.

1.1 Reagents: Magnesate(1-), dichloro(1-methylethyl)-, lithium (1:1)

Solvents: Tetrahydrofuran; 3 h, -20 °C

1.2 Solvents: Toluene; 3 h, 23 °C

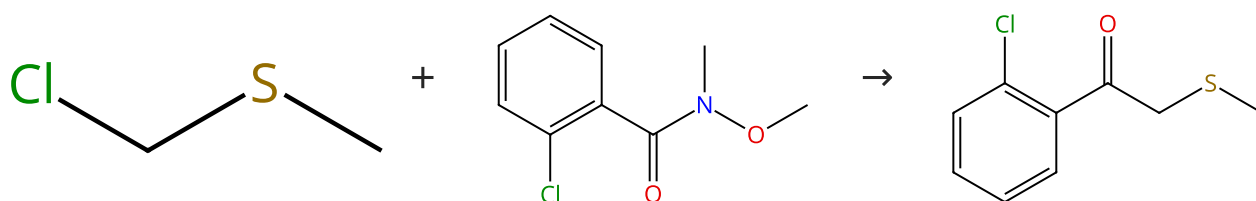
1.3 Reagents: Ammonium chloride

Solvents: Water

Experimental Protocols

Scheme 32 (1 Reaction)

Steps: 1 Yield: 94%



Suppliers (38)

Suppliers (54)

Suppliers (17)

31-373-CAS-18866859

Steps: 1 Yield: 94%

Substituted α-Sulfur Methyl Carbanions: Effective Homologating Agents for the Chemoselective Preparation of β-Oxo Thioethers from Weinreb Amides

By: Senatore, Raffaele; et al

European Journal of Organic Chemistry (2018), 2018(20-21), 2466-2470.

1.1 Reagents: Lithium
Catalysts: 4,4'-Bis(1,1-dimethylethyl)-1,1'-biphenyl

Solvents: Tetrahydrofuran; -78 °C

1.2 Solvents: Tetrahydrofuran; 3 h, -78 °C

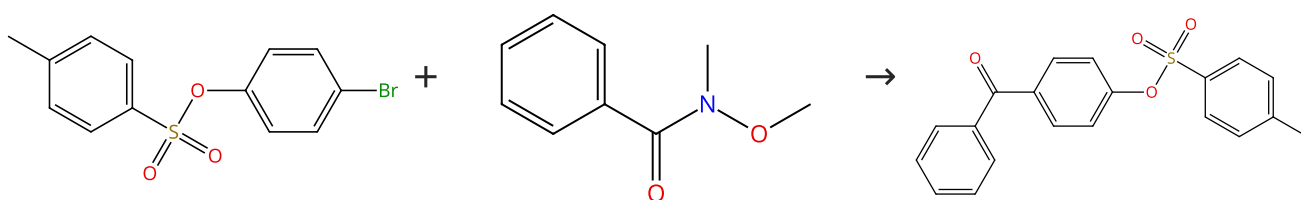
1.3 Reagents: Ammonium chloride

Solvents: Water

Experimental Protocols

Scheme 33 (1 Reaction)

Steps: 1 Yield: 93%

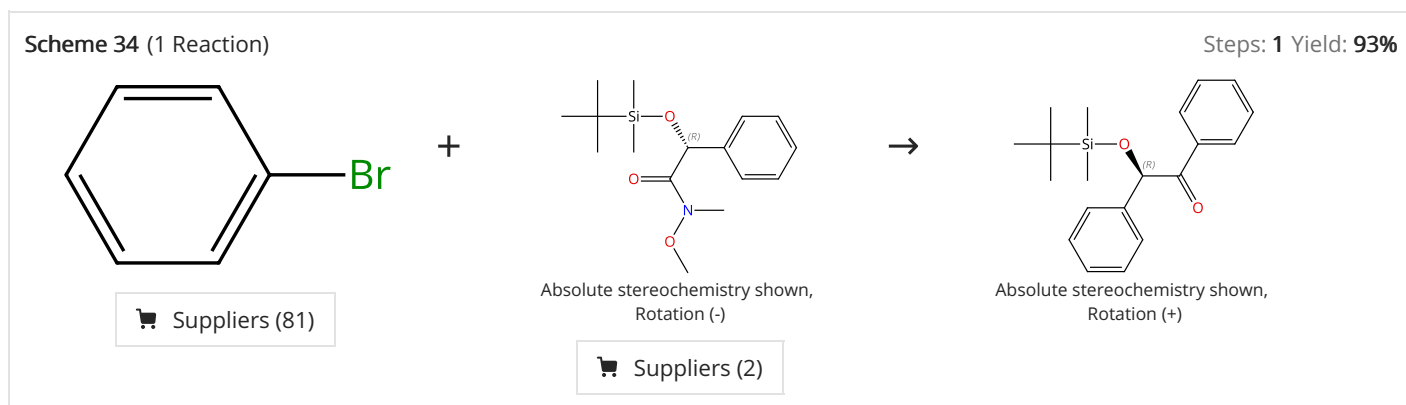


Suppliers (25)

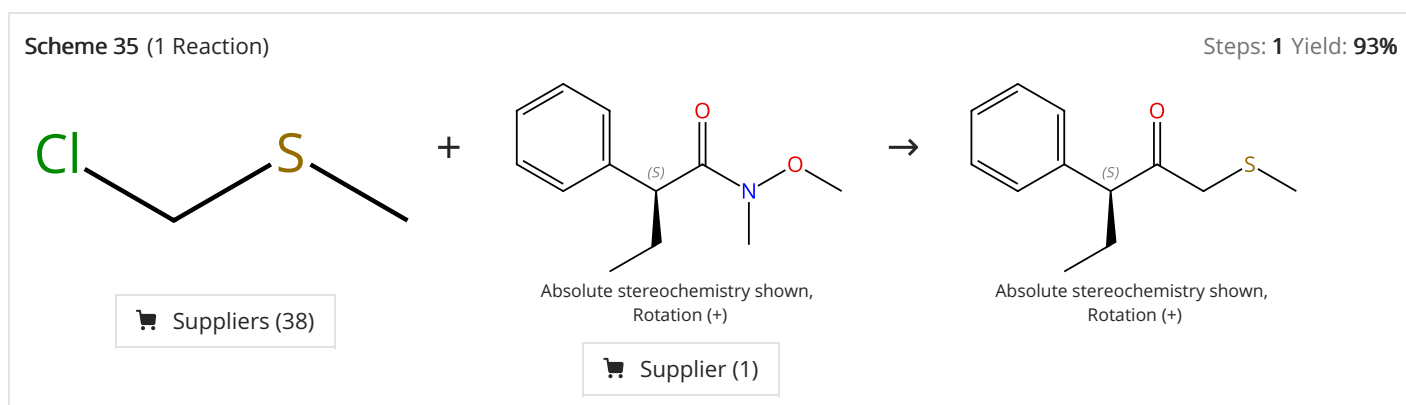
Suppliers (72)

Suppliers (3)

31-373-CAS-22198672 Steps: 1 Yield: 93% 1.1 Reagents: Magnesate(1-), dichloro(1-methylethyl)-, lithium (1: 1) Solvents: Tetrahydrofuran; 36 h, 0 °C 1.2 Solvents: Toluene; 3 h, 23 °C 1.3 Reagents: Ammonium chloride Solvents: Water Experimental Protocols	Synthesis of biaryl ketones by arylation of Weinreb amides with functionalized Grignard reagents under thermodynamic control vs. kinetic control of N,N-Boc₂-amides By: Li, Guangchen; et al Organic & Biomolecular Chemistry (2020), 18(20), 3827-3831.
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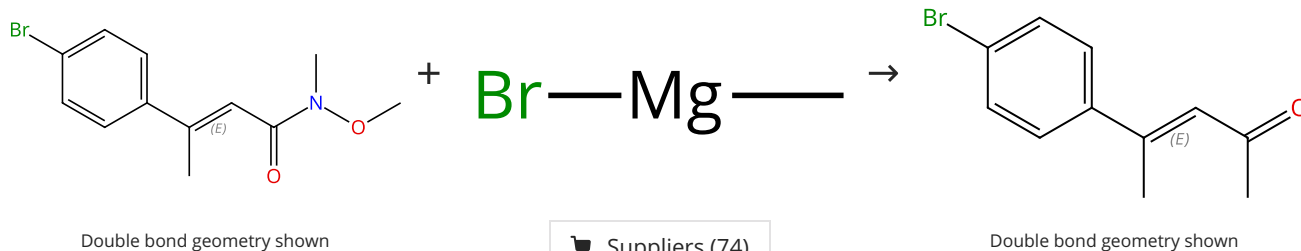
31-614-CAS-32738522 Steps: 1 Yield: 93% 1.1 Reagents: Butyllithium Solvents: Tetrahydrofuran, Hexane; -78 °C; 1 h, -78 °C 1.2 Solvents: Tetrahydrofuran; -78 °C; 2.5 - 3 h, -78 °C 1.3 Reagents: Ammonium chloride Solvents: Water Experimental Protocols	Accessing Diverse Cross-Benzoin and α-Siloxy Ketone Products via Acyl Substitution Chemistry By: Larcombe, Chloe N.; et al Journal of Organic Chemistry (2022), 87(14), 9408-9413.
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31-373-CAS-18866849 Steps: 1 Yield: 93% 1.1 Reagents: Lithium Catalysts: 4,4'-Bis(1,1-dimethylethyl)-1,1'-biphenyl Solvents: Tetrahydrofuran; -78 °C 1.2 Solvents: Tetrahydrofuran; 3 h, -78 °C 1.3 Reagents: Ammonium chloride Solvents: Water Experimental Protocols	Substituted α-Sulfur Methyl Carbanions: Effective Homologating Agents for the Chemoselective Preparation of β-Oxo Thioethers from Weinreb Amides By: Senatore, Raffaele; et al European Journal of Organic Chemistry (2018), 2018(20-21), 2466-2470.
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Scheme 36 (1 Reaction)

Steps: 1 Yield: 93%



Suppliers (74)

Suppliers (3)

31-373-CAS-22257382

Steps: 1 Yield: 93%

1.1 Solvents: Tetrahydrofuran; -30 °C; -30 °C → -5 °C; 0.5 h, -5 °C

1.2 Reagents: Ammonium chloride
Solvents: Water

Experimental Protocols

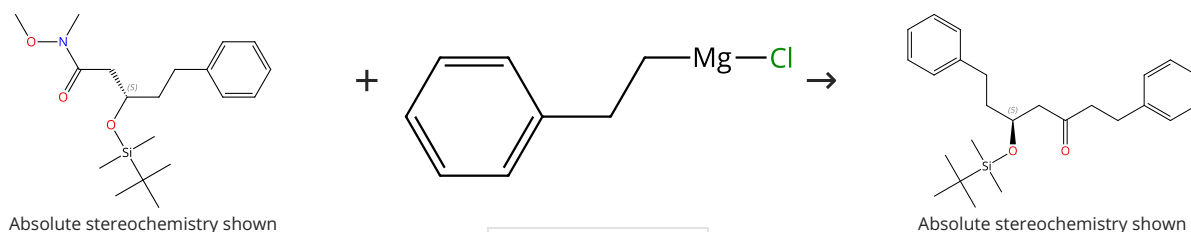
The synthesis of non-racemic β -alkyl- β -aryl-disubstituted allyl alcohols and their transformation into bearing a quaternary stereocenter allylamines and amino acids

By: Narczyk, Aleksandra; et al

Organic & Biomolecular Chemistry (2018), 16(21), 3921-3946.

Scheme 37 (1 Reaction)

Steps: 1 Yield: 93%



Suppliers (15)

31-614-CAS-44581393

Steps: 1 Yield: 93%

1.1 Solvents: Tetrahydrofuran; -78 °C; 30 min, -78 °C

Experimental Protocols

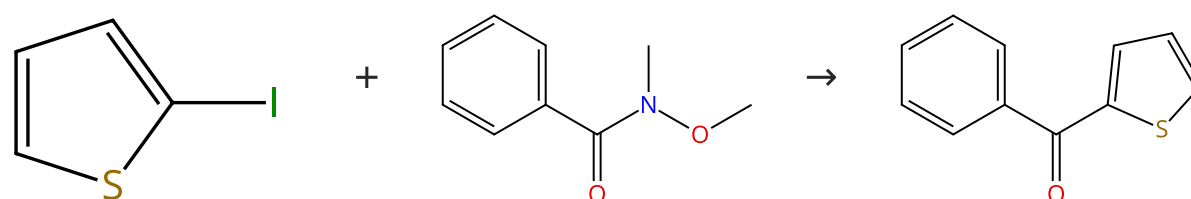
Practical Preparation of (3S)-Hydroxy-5-Phenylpentanoic Acid: Asymmetric Synthesis of (S)-Daphneolone and (S)-Dihydroxy ashabushiketol, and Formal Synthesis of (3S,5S)-Yashabu shidiol B

By: Nam, So-Yeon; et al

International Journal of Molecular Sciences (2025), 26(4), 1476.

Scheme 38 (1 Reaction)

Steps: 1 Yield: 93%



Suppliers (78)

Suppliers (72)

Suppliers (71)

31-373-CAS-22198682

Steps: 1 Yield: 93%

1.1 Reagents: Magnesate(1-), dichloro(1-methylethyl)-, lithium (1: 1)

Solvents: Tetrahydrofuran; 30 min, 23 °C

1.2 Solvents: Toluene; 3 h, 23 °C

1.3 Reagents: Ammonium chloride
Solvents: Water

Experimental Protocols

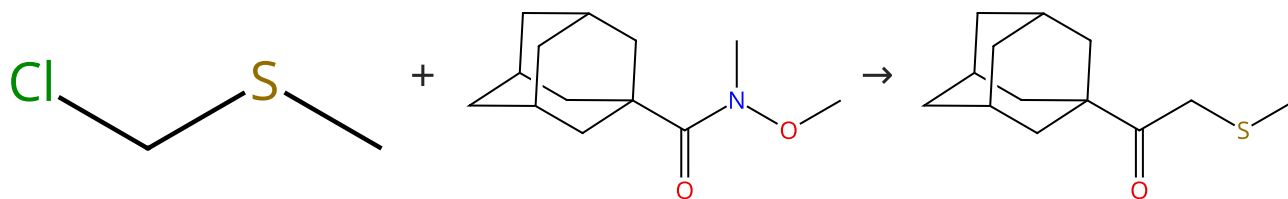
Synthesis of biaryl ketones by arylation of Weinreb amides with functionalized Grignard reagents under thermodynamic control vs. kinetic control of N,N-Boc₂-amides

By: Li, Guangchen; et al

Organic & Biomolecular Chemistry (2020), 18(20), 3827-3831.

Scheme 39 (1 Reaction)

Steps: 1 Yield: 92%



Suppliers (38)

Suppliers (21)

Suppliers (5)

31-373-CAS-18866865

Steps: 1 Yield: 92%

Substituted α -Sulfur Methyl Carbanions: Effective Homologating Agents for the Chemoselective Preparation of β -Oxo Thioethers from Weinreb Amides

By: Senatore, Raffaele; et al

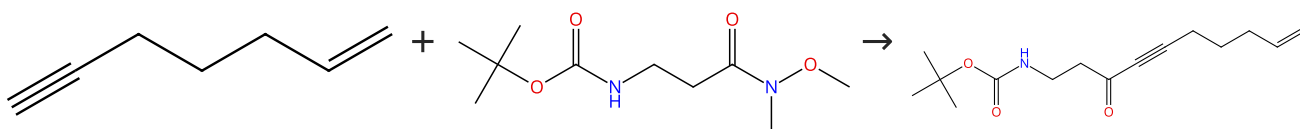
European Journal of Organic Chemistry (2018), 2018(20-21), 2466-2470.

- 1.1 **Reagents:** Lithium
Catalysts: 4,4'-Bis(1,1-dimethylethyl)-1,1'-biphenyl
Solvents: Tetrahydrofuran; -78 °C
- 1.2 **Solvents:** Tetrahydrofuran; 3 h, -78 °C
- 1.3 **Reagents:** Ammonium chloride
Solvents: Water

Experimental Protocols

Scheme 40 (1 Reaction)

Steps: 1 Yield: 92%



Suppliers (25)

Suppliers (52)

31-614-CAS-36832585

Steps: 1 Yield: 92%

Accessing spiropiperidines from dihydropyridones through tandem triflation-allylation and ring-closing metathesis (RCM)

By: Gantasala, Naresh; et al

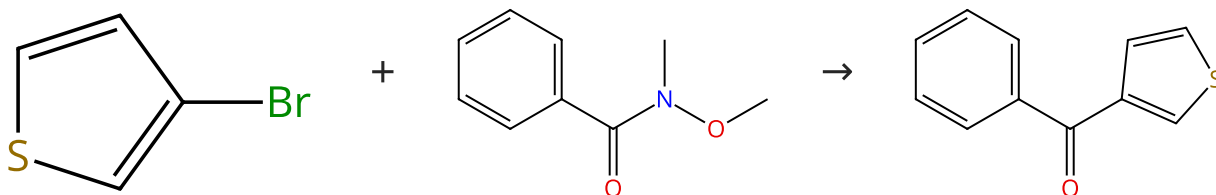
Organic & Biomolecular Chemistry (2023), 21(25), 5245-5253.

- 1.1 **Reagents:** Butyllithium
Solvents: Tetrahydrofuran, Hexane; -78 °C; 1 h, -78 °C
- 1.2 **Solvents:** Tetrahydrofuran; -78 °C; 15 min, -78 °C; 1 h, -50 °C; -50 °C $\rightarrow$ -5 °C; 1.5 h, -5 °C
- 1.3 **Reagents:** Monosodium phosphate
Solvents: Water

Experimental Protocols

Scheme 41 (1 Reaction)

Steps: 1 Yield: 92%

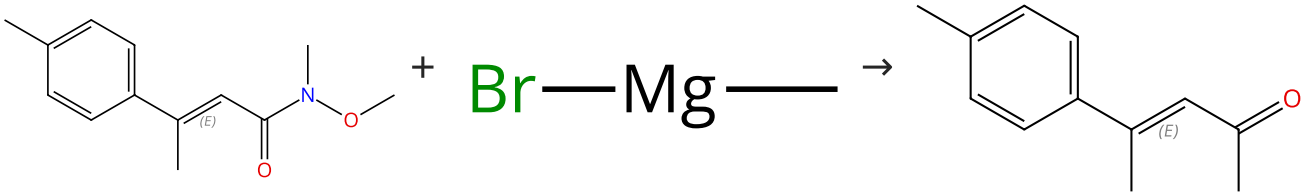


Suppliers (81)

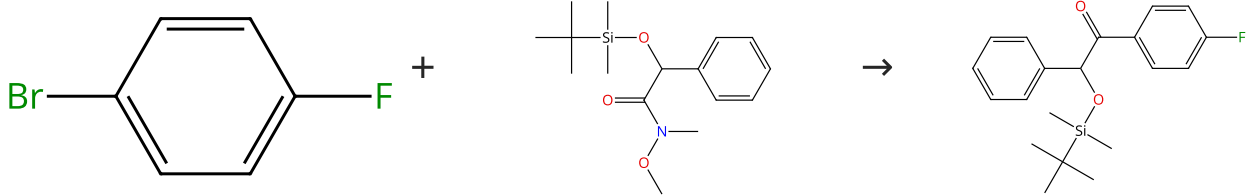
Suppliers (72)

Suppliers (31)

31-373-CAS-22198683 Steps: 1 Yield: 92% 1.1 Reagents: Magnesate(1-), dichloro(1-methylethyl)-, lithium (1:1) Solvents: Tetrahydrofuran; 30 min, 23 °C 1.2 Solvents: Toluene; 3 h, 23 °C 1.3 Reagents: Ammonium chloride Solvents: Water	Synthesis of biaryl ketones by arylation of Weinreb amides with functionalized Grignard reagents under thermodynamic control vs. kinetic control of N,N-Boc₂-amides By: Li, Guangchen; et al Organic & Biomolecular Chemistry (2020), 18(20), 3827-3831.
Experimental Protocols	

Scheme 42 (1 Reaction)  Double bond geometry shown Supplier (1)	Steps: 1 Yield: 92% Suppliers (74) Double bond geometry shown Suppliers (3)
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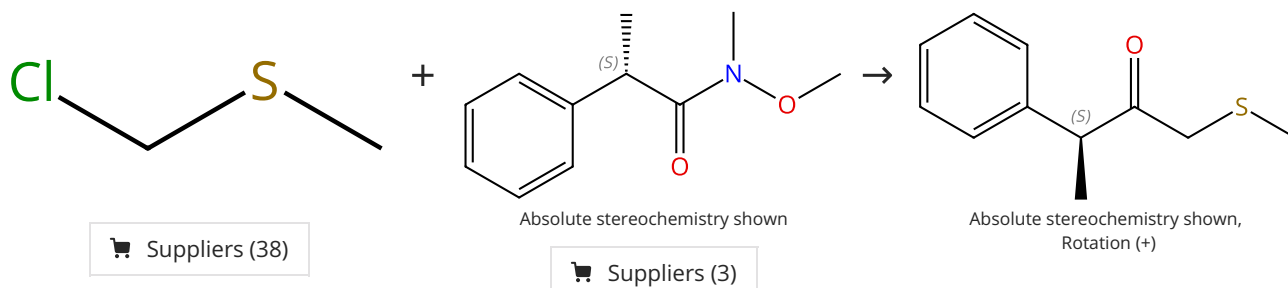
31-373-CAS-18800524 Steps: 1 Yield: 92% 1.1 Solvents: Tetrahydrofuran; -30 °C; -30 °C → -5 °C; 0.5 h, -5 °C 1.2 Reagents: Ammonium chloride Solvents: Water	The synthesis of non-racemic β-alkyl-β-aryl-disubstituted allyl alcohols and their transformation into bearing a quaternary stereocenter allylamines and amino acids By: Narczyk, Aleksandra; et al
Experimental Protocols	Organic & Biomolecular Chemistry (2018), 16(21), 3921-3946.

Scheme 43 (1 Reaction)  Suppliers (103) Supplier (1)	Steps: 1 Yield: 92%
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31-614-CAS-32738505 Steps: 1 Yield: 92% 1.1 Reagents: Butyllithium Solvents: Tetrahydrofuran, Hexane; -78 °C; 1 h, -78 °C 1.2 Solvents: Tetrahydrofuran; -78 °C; 2.5 - 3 h, -78 °C 1.3 Reagents: Ammonium chloride Solvents: Water	Accessing Diverse Cross-Benzoin and α-Siloxy Ketone Products via Acyl Substitution Chemistry By: Larcombe, Chloe N.; et al Journal of Organic Chemistry (2022), 87(14), 9408-9413.
Experimental Protocols	

Scheme 44 (1 Reaction)

Steps: 1 Yield: 91%



31-373-CAS-18866851

Steps: 1 Yield: 91%

Substituted α -Sulfur Methyl Carbanions: Effective Homologating Agents for the Chemoselective Preparation of β -Oxo Thioethers from Weinreb Amides

By: Senatore, Raffaele; et al

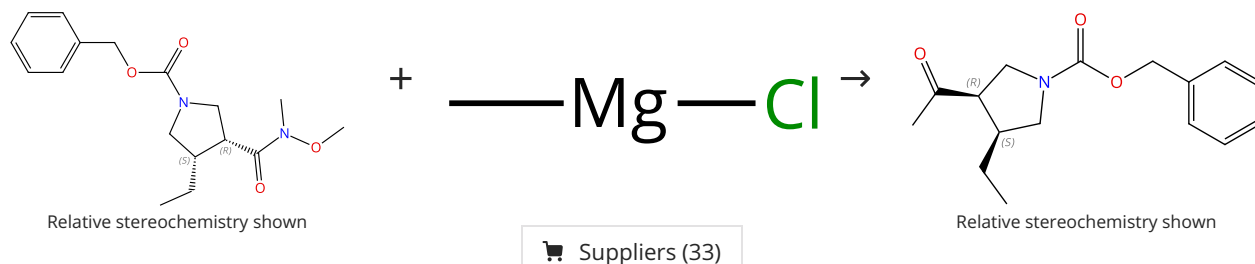
European Journal of Organic Chemistry (2018), 2018(20-21), 2466-2470.

- 1.1 Reagents: Lithium
Catalysts: 4,4'-Bis(1,1-dimethylethyl)-1,1'-biphenyl
Solvents: Tetrahydrofuran; -78 °C
- 1.2 Solvents: Tetrahydrofuran; 3 h, -78 °C
- 1.3 Reagents: Ammonium chloride
Solvents: Water

Experimental Protocols

Scheme 45 (1 Reaction)

Steps: 1 Yield: 91%



31-373-CAS-23798347

Steps: 1 Yield: 91%

Process preparation of Upadacitinib and useful intermediates via Weinreb amides

Assignee: Crystal Pharma, S.A.U.

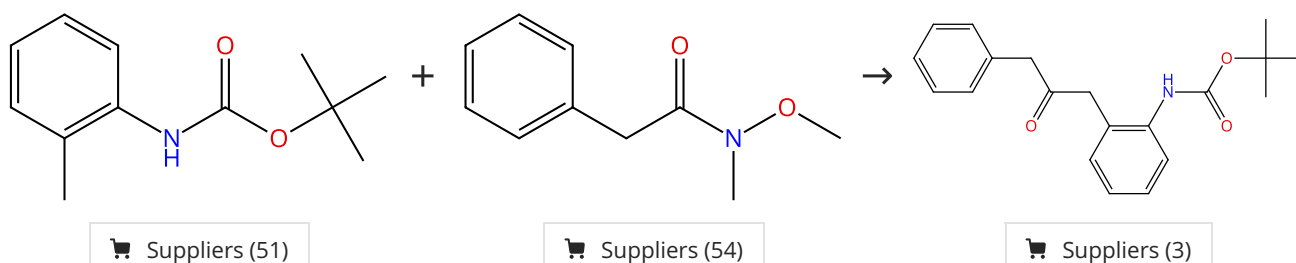
World Intellectual Property Organization, WO2021123288 A1 2021-06-24

[PatentPak](#) available

- 1.1 Solvents: Tetrahydrofuran; 0 - 5 °C; 15 - 30 min, 0 - 5 °C
- 1.2 Reagents: Ammonium chloride
Solvents: Water

Scheme 46 (1 Reaction)

Steps: 1 Yield: 91%



31-373-CAS-21236794

Steps: 1 Yield: 91%

Facile synthesis of chiral indolines through asymmetric hydrogenation of in situ generated indoles

By: Yu, Chang-Bin; et al

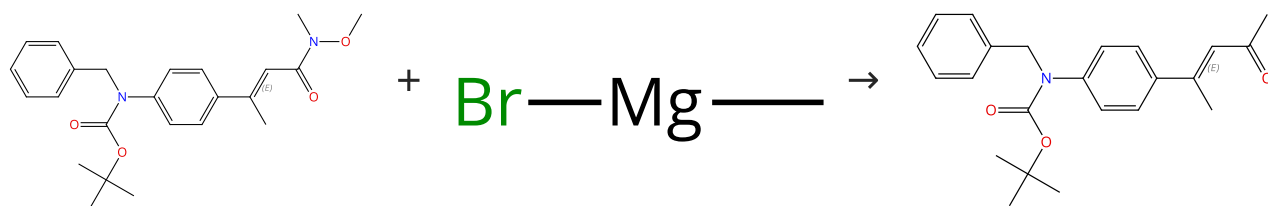
Organic Chemistry Frontiers (2018), 5(19), 2805-2809.

- 1.1 Reagents: *sec*-Butyllithium
Solvents: Tetrahydrofuran, Hexane; -40 °C; 1 h, -40 °C
- 1.2 Solvents: Tetrahydrofuran; 30 min, -40 °C
- 1.3 Reagents: Water

Experimental Protocols

Scheme 47 (1 Reaction)

Steps: 1 Yield: 91%



Double bond geometry shown

Suppliers (74)

Double bond geometry shown

31-373-CAS-18800527

Steps: 1 Yield: 91%

1.1 Solvents: Tetrahydrofuran; -30 °C; -30 °C → -5 °C; 0.5 h, -5 °C

1.2 Reagents: Ammonium chloride
Solvents: Water

Experimental Protocols

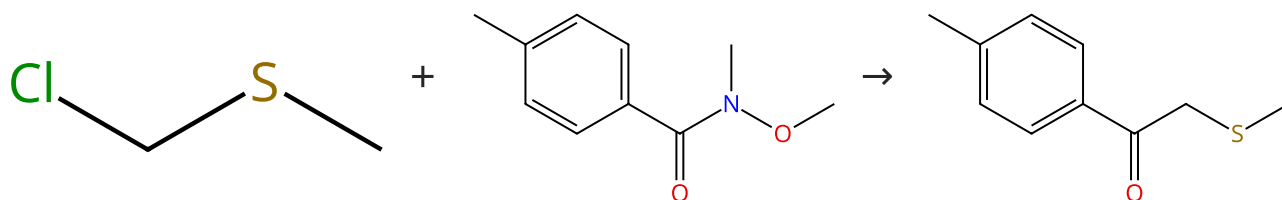
The synthesis of non-racemic β -alkyl- β -aryl-disubstituted allyl alcohols and their transformation into bearing a quaternary stereocenter allylamines and amino acids

By: Narczyk, Aleksandra; et al

Organic & Biomolecular Chemistry (2018), 16(21), 3921-3946.

Scheme 48 (1 Reaction)

Steps: 1 Yield: 91%



Suppliers (38)

Suppliers (60)

Suppliers (15)

31-373-CAS-18866854

Steps: 1 Yield: 91%

1.1 Reagents: Lithium
Catalysts: 4,4'-Bis(1,1-dimethylethyl)-1,1'-biphenyl
Solvents: Tetrahydrofuran; -78 °C

1.2 Solvents: Tetrahydrofuran; 3 h, -78 °C

1.3 Reagents: Ammonium chloride
Solvents: Water

Experimental Protocols

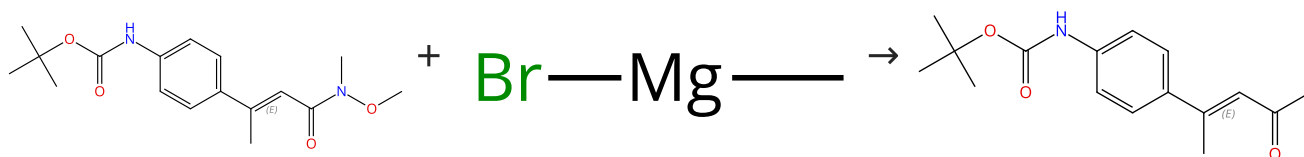
Substituted α -Sulfur Methyl Carbanions: Effective Homologating Agents for the Chemoselective Preparation of β -Oxo Thioethers from Weinreb Amides

By: Senatore, Raffaele; et al

European Journal of Organic Chemistry (2018), 2018(20-21), 2466-2470.

Scheme 49 (1 Reaction)

Steps: 1 Yield: 91%



Double bond geometry shown

Suppliers (74)

Double bond geometry shown

31-373-CAS-18800526

Steps: 1 Yield: 91%

1.1 Solvents: Tetrahydrofuran; -30 °C; -30 °C → -5 °C; 0.5 h, -5 °C

1.2 Reagents: Ammonium chloride
Solvents: Water

Experimental Protocols

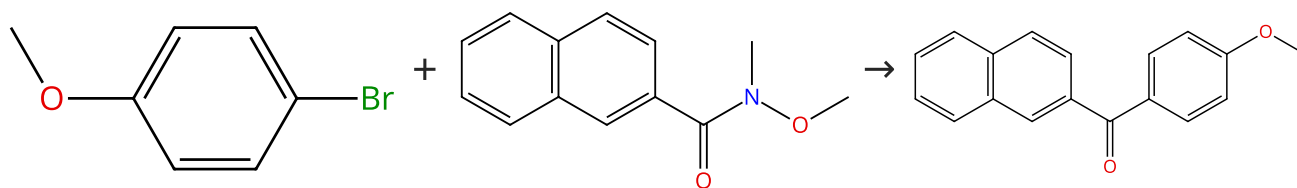
The synthesis of non-racemic β -alkyl- β -aryl-disubstituted allyl alcohols and their transformation into bearing a quaternary stereocenter allylamines and amino acids

By: Narczyk, Aleksandra; et al

Organic & Biomolecular Chemistry (2018), 16(21), 3921-3946.

Scheme 50 (1 Reaction)

Steps: 1 Yield: 90%



Suppliers (71)

Suppliers (53)

Suppliers (12)

31-373-CAS-22198691

Steps: 1 Yield: 90%

1.1 **Reagents:** Magnesate(1-), dichloro(1-methylethyl)-, lithium (1:1)**Solvents:** Tetrahydrofuran; 3 d, 23 °C1.2 **Solvents:** Toluene; 3 h, 23 °C1.3 **Reagents:** Ammonium chloride**Solvents:** Water

Experimental Protocols

Synthesis of biaryl ketones by arylation of Weinreb amides with functionalized Grignard reagents under thermodynamic control vs. kinetic control of N,N-Boc₂-amides

By: Li, Guangchen; et al

Organic & Biomolecular Chemistry (2020), 18(20), 3827-3831.